

Development of a 3-RRS Parallel Manipulator for Automated Intra-vehicle Liquid Control of a Coffee Cup

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Abstract: Transportation of liquid poses a challenge in many industries owing to the free movement of the center of mass of the body, and the low viscosity of many substances. Research into liquid control spans from medical applications, such as damping the oscillations of drinking water for Parkinson's patients, to large-scale industrial processes, such as transporting smelted materials. In all cases, rapid system response and corrective action is needed to prevent spillage. In this paper, an autonomous 3-Revolute-Revolute-Spherical (3-RRS) Parallel Manipulator is proposed, with servo-motor actuation, and micro-electromechanical (MEMS) sensor fusion, for effective liquid control of a cup of coffee in a moving vehicle. To realise the system, a Complementary Filter (CF) sensor fusion algorithm, and trajectory tracking of the liquid's center of mass, is used to estimate the body's orientation in space. Additionally, a proportional-integral-derivative (PID) controller is designed for servomotor actuation and disturbance rejection. Finally, inverse kinematic (IK) computation is performed to manipulate a custom-designed 3-legged parallel manipulator. The effectiveness of the coffee stabilisation system in preventing liquid spillage is demonstrated under field conditions by braking, accelerating, and turning a moving vehicle.

Keywords: liquid control; autonomous systems; parallel manipulator; sensor fusion; orientation estimation; inverse kinematics

1. Introduction

Autonomous systems have a growing presence in a variety of industrial and domestic sectors, with many tasks that are complex or dangerous for humans to perform often being completed with robotic assistance. Liquid sloshing-suppression is one such area where human involvement has shortcomings. These systems hold great importance for a variety of fields, such as beverage manufacturing, fuel-tank transportation, and water-tank design against earthquake vibrations [1]. This paper proposes the design of an automated intra-vehicle coffee stabiliser and anti-spill device using a 3-RRS parallel manipulator, which demonstrates the use of precise liquid control in a mobile environment.

To realise the system, three main elements needed to be considered: a sensor suite, a mechanical structure for actuation, and a control subsystem. In terms of the sensor suite, two approaches could be taken to capture the environmental changes imposed on the coffee cup: deduction based on the liquid's changing motion, or direct measurement of the applied forces on the system. In terms of the former, one such implementation is a computer vision system using a camera and a machine-learning model to classify the orientation of the coffee in the cup [2]. Another is to use mm-wave radar to identify changes in the composition of the liquid's surface [3]. These implementations rely on

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sensors located at different positions on the vehicle, away from the cup itself, and are susceptible to interference from the driver and passengers, as well as processing delays. Alternatively, in the case of the latter, the linear and angular accelerations applied to the system could be directly measured using an inertial measurement unit (IMU) [4]. Raw IMU sensor data contains bias and jitter, requiring sensor fusion and filtering for smoother results, as explained by Gui et. al [5]. There are several filtering options for this direct approach, such as the Kalman Filter (KF) [6], the Estimation by Gradient Descent (EGD) algorithm proposed by Magdwick et. al [7], and the Complimentary Filter (CF) [5], each suited for different applications.

According to the mechanical structure, the type of robotic architecture employed depends on the system application and its physical constraints. For a coffee-cup actuator, either a serial manipulator, such as a 3-axis planar robot [4], or a parallel manipulator, such as a 3-Revolute-Prismatic-Spherical (3-RPS) parallel manipulator [8], would be suitable for liquid control. The 3-axis platform robotic manipulator in the gimbal structure shown by Rafiq et. al [4] offers independent roll and pitch motions that when combined, achieve the desired kinematic orientation. Planar manipulators are mathematically simpler, making for easier control design, as the total linear and angular acceleration is the sum of the accelerations along each individual axis [4], however, these serial designs lack the same rigidity and carrying capacity. In contrast, parallel manipulators are constrained holonomic robots, which pose a challenge for deriving the dynamic model and IK of the system. For model design, Lagrangian [9], geometric [10], and screw theory methods [8] offer derivation processes for motion planning of parallel manipulators.

In terms of controller design, the coffee-cup stabiliser can be classified as either whole-body control, or as decoupled control. The high-dimensionality of parallel manipulators, in conjunction with the slosh model of liquid, makes whole-body control very complex. To simplify the process, trajectory tracking can be employed by modelling the liquid coffee as a pendulum [13], and two decoupled control systems can be designed - one for each axis. Several controller architectures are available for implementation, each suited for different applications. Sliding-mode control (SMC) offers a fast response time and effective disturbance rejection [9], however to achieve the set-point targets, powerful motor action is needed. Fuzzy Logic (FL) implementation is another method for non-linear systems, using 'if-then' logic to map 'crisp' input values to membership values [14]. Another design commonly used in IMU-based actuation systems is the PID controller [11]. The PID algorithm offers a trade-off between speed and stability. When paired with an orientation estimation filter design, it offers an effective controller method, as shown by Popelka [12].

The final system design included a direct IMU implementation, combined with a CF sensor fusion algorithm to achieve an accurate and efficient classification of the applied forces. A 3-RRS parallel manipulator with servo-motor actuation, and PID control architecture, was employed to manipulate the mechanical structure. The system was tested in a moving vehicle, applying turning, braking, and accelerating motions, and was able to prevent liquid spillage. The remaining parts of the paper are organised as follows: Section 2 reviews related literature on liquid control and parallel manipulators. Section 3 gives an overview of the system design and implementation. Section 4 presents the experimental results, whereby the system was able to track the target resultant roll and pitch vector with an absolute average roll and pitch error of 0.7162° , and 0.7633° , respectively. Lastly, Section 5 offers a discussion into the results, and Section 6 concludes the work.

2. Related Work

There are several studies conducted on liquid control and sloshing dynamics, as well as research into mechanical structures for oscillation damping in fluid systems, however, few studies have been conducted in the environmental context presented in this paper.

Vidya et. al [11] considers the shortcomings of liquid containers for individuals with disability impediments. A wearable auto-stabilising cup holder is proposed for Parkinson's Disease patients. The mechanical structure was designed as a serial manipulator, using a two degree-of-freedom (DOF) structure actuated by servo motors. An IMU sensor was used for vibration detection, and a PID controller was implemented to minimise the oscillations in the cup. This simple approach offers a cost-effective method for reducing the unbound fluid motions that arise from jittery hand movements.

In an industrial application, Yano et. al [1] demonstrate the use of H^∞ control for complete sloshing suppression in the container transfer of smelted materials. The paper proposes a robust control system paired with an active transfer machine that incorporates 2-DOF. A nominal model was designed, aimed at the middle value of the natural frequency of sloshing, thereby suppressing its effect on the system. The results were compared to linear quadratic integration (LQI) control design, and shown to be more effective in preventing residual vibrations, however for complete suppression, an active mechanism was needed in conjunction with the H^∞ controller. When combined, the research was able to effectively suppress sloshing during the acceleration and deceleration phases of the transfer.

Similarly, Biagiotti et. al propose the use of a feed-forward control implementation for damping liquid oscillations during motion [13]. A plug-in feed-forward controller was used in the form of a damped harmonic smoother between the motion planner of the system, and the robotic controller. The smoother was composed of a two-impulse input-shaper and a second-order low pass filter (LPF), which suppressed higher sloshing modes of the liquid. A tilting-compensation algorithm was also needed to counteract the lateral accelerations of the system. This method aimed to remove *a priori* knowledge of the reference trajectory, however the correct estimation of certain parameters, such as the natural frequency and damping ratio of the liquid, was needed for effective system performance.

Active Acceleration Compensation during high-speed transfer, proposed by Chen et. al [15] is another approach for liquid control. The system employed the "Waiter-Tray" model, which adapts the orientation of the robot's end-effector so that sloshing effects are minimised. This approach aimed to reduce the "critical switching-zones" which occur when acceleration abruptly changes during robotic transfer, and a relative motion is created between the liquid and the body. An optimisation filter was applied to find the ideal tilting angles, diminishing the oscillations from the motion, and thus a time-optimal approach was achieved.

Overall, existing liquid control methods have advantages and shortcomings, with many systems requiring both a control algorithm, and an active mechanism for effective slosh suppression. Continued research into this field is critical for safer and more efficient liquid-transportation and vibration-damping methods, ranging from domestic use-cases to industrial applications. This paper aims to further the body of knowledge in liquid control by adopting a 3-RRS parallel manipulator design in the context of an anti-spill system for a cup of coffee in a moving vehicle.

3. Materials and Methods

This section details the approach, design and integration of each subsystem in the coffee-stabiliser, as well as an overview of the software and hardware components.

3.1. Design Approach

For liquid control in a moving vehicle, the system should be able to react to the changing accelerations that are typically experienced during normal driving conditions. This includes disturbances on the system from mechanical vibrations, uneven road surfaces, and driving motions. Assuming the cup is filled to the top (at a minimum 1 cm from the cup rim), the center of the body coordinate system (x_b, y_b, z_b) is fixed at the center of the liquid surface, as shown in Figure 1. When an applied acceleration (a_{app}) occurs on the global coordinate system (x_G, y_G, z_G), fixed at the base of the coffee-cup stabiliser, a resultant acceleration (a_{res}) occurs. In order to prevent spillage, the system vector (a_{sys}), with corresponding roll (φ_r) and pitch (θ_p) angles, needs to be maintained. The vector is approximately perpendicular to the base of the cup and is equal in magnitude but opposite in direction to the resultant acceleration.

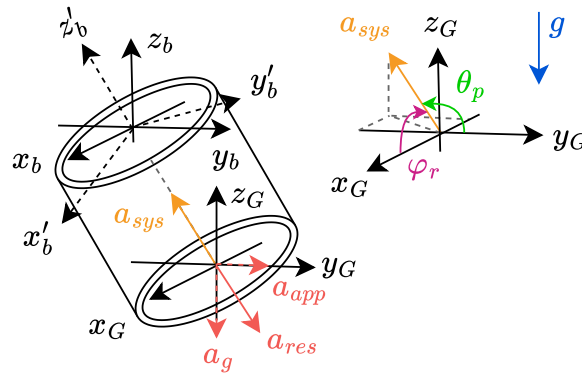


Figure 1. Overview of system approach to designing the coffee-cup stabiliser

To achieve the system objective, the functional block diagram in Figure 2 shows the high-level system approach for slosh suppression.

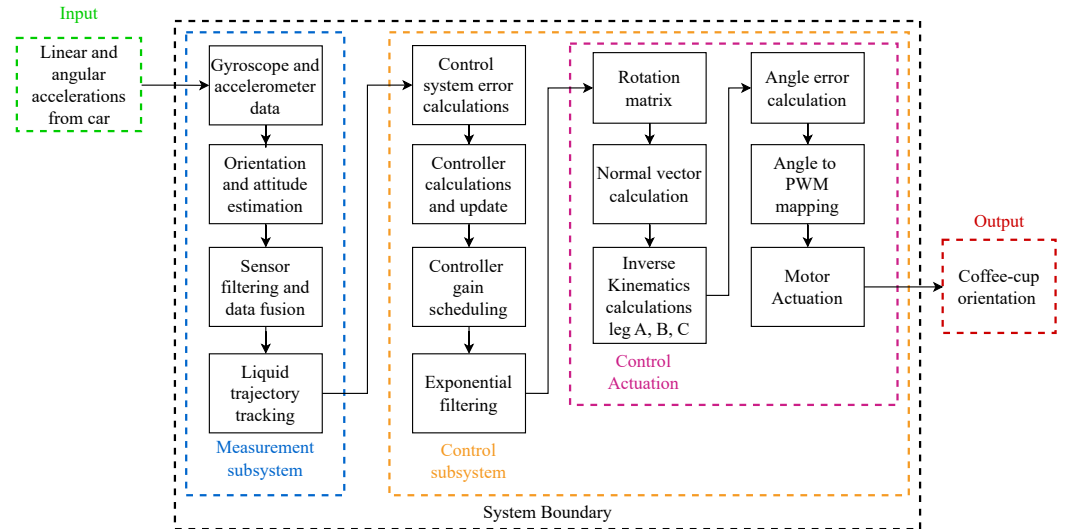


Figure 2. System flow diagram

3.2. Measurement Subsystem

An IMU, containing a gyroscope and accelerometer, as part of the internal MEMS sensor, is needed to capture the linear and angular accelerations applied to the system. Raw sensor data is susceptible to errors, as the accelerometer contains noise from the rapid fluctuations of the IMU mechanics, and the gyroscope contains drift from accumulated

errors over time. This is demonstrated by simulating the rotation of an LSM6DSOX IMU [22] in MATLAB. Converting the Euler angle roll, pitch, and yaw sequence into quaternions,

$$\begin{bmatrix} q_0 \\ q_1 \\ q_2 \\ q_3 \end{bmatrix} = \begin{bmatrix} C\varphi C\theta C\psi + S\varphi S\theta S\psi \\ -C\varphi S\theta S\psi + S\varphi C\theta C\psi \\ C\varphi S\theta C\psi + S\varphi C\theta S\psi \\ C\varphi C\theta S\psi - S\varphi S\theta C\psi \end{bmatrix}, \quad (1)$$

where $C\varphi \triangleq \cos(\varphi)$ and $S\varphi \triangleq \sin(\varphi)$, and the orientation quaternion is given as $q = [q_0 \ q_1 \ q_2 \ q_3]$; applying the Direct Cosine Matrix (DCM),

$$DCM = \begin{bmatrix} (q_0^2 + q_1^2 - q_2^2 - q_3^2) & 2(q_1q_2 + q_0q_3) & 2(q_1q_3 - q_0q_2) \\ 2(q_1q_2 - q_0q_3) & (q_0^2 - q_1^2 + q_2^2 - q_3^2) & 2(q_2q_3 + q_0q_1) \\ 2(q_1q_3 + q_0q_2) & 2(q_2q_3 + q_0q_1) & (q_0^2 - q_1^2 - q_2^2 + q_3^2) \end{bmatrix}, \quad (2)$$

to perform the rotation sequence of the quaternion, the accelerometer and gyroscope can be found as

$$\begin{aligned} v &= \begin{bmatrix} 0 & 0 & -9.807 \end{bmatrix}^T / (-9.807), \\ a &= DCM \times v. \end{aligned} \quad (3)$$

To view the simulated results, the quaternion is converted back to Euler angles as

$$\begin{bmatrix} \theta \\ \varphi \\ \psi \end{bmatrix} = \begin{bmatrix} \arcsin(-2(q_1q_2 - q_0q_3)) \\ \arctan(2(q_0q_1 + q_2q_3) / (q_0^2 - q_1^2 - q_2^2 + q_3^2)) \\ \arctan(2(q_0q_3 + q_1q_2) / (q_0^2 + q_1^2 - q_2^2 - q_3^2)) \end{bmatrix}. \quad (4)$$

From the rotation sequence, the MATLAB simulation in Figure 3 shows the effects of the IMU noise on the sensor output.

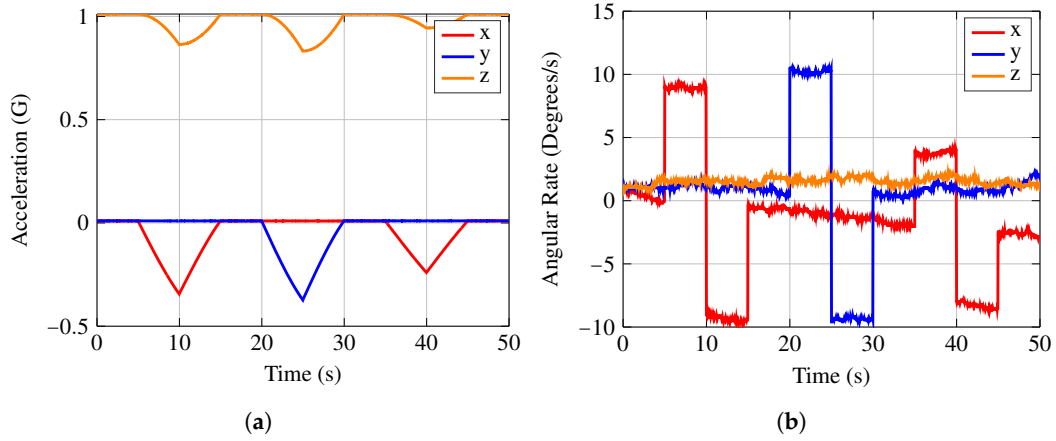


Figure 3. (a) Accelerometer rotation sequence with jitter. (b) Gyroscope rotation sequence with drift.

To accurately estimate the orientation of the cup's position in space, sensor fusion is needed. The CF shown in Figure 4 and Table 1 combines the gyroscope and accelerometer data, using a HPF to remove the gyroscope drift, and a LPF to remove the accelerometer jitter.

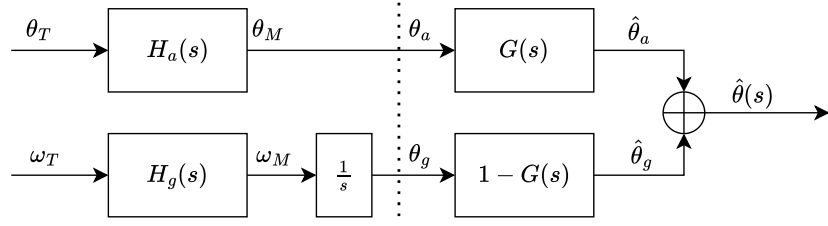


Figure 4. CF algorithm block diagram, taken from Gui. et. al [5]

Table 1. CF symbol meanings.

Symbol	Meaning	Symbol	Meaning
θ_T	Ground truth linear acceleration	ω_T	Ground truth angular acceleration
θ_M	Accelerometer data with jitter	ω_M	Gyroscope data with bias
H_a	Accelerometer transfer function	H_g	Gyroscope transfer function
θ_a	Acceleration angle	θ_g	Gyroscope angle
$G(s)$	Low-pass filter	$1 - G(s)$	High-pass filter
$\hat{\theta}(s)$	Orientation angle		

From the accelerometer, the tilt angle is computed as

$$\theta_a = \begin{bmatrix} \theta_x \\ \theta_y \\ \theta_z \end{bmatrix} = \begin{bmatrix} \arctan(a_y/a_z) \\ \arctan(-a_x/\sqrt{a_y^2 + a_z^2}) \\ 0 \end{bmatrix}. \quad (5)$$

Similarly, the orientation can be estimated using the gyroscope as,

$$\theta_\omega = \begin{bmatrix} \theta_{xt} \\ \theta_{yt} \\ \theta_{zt} \end{bmatrix} = \begin{bmatrix} \theta_{xt-1} + \omega_x \Delta_t \\ \theta_{yt-1} + \omega_y \Delta_t \\ \theta_{zt-1} + \omega_z \Delta_t \end{bmatrix}, \quad (6)$$

where Δ_t is the sampling time of the IMU [19]. Combining these equations, the orientation in terms of pitch and roll is found as

$$\hat{\theta} = \alpha \theta_\omega + (1 - \alpha) \theta_a. \quad (7)$$

The controlling parameter α is a value between $[0, 1]$, and determines the filter gain, limiting how much the gyroscope and the accelerometer each contribute to the final orientation estimation. The adjustable filter parameter is set to $\alpha = 0.96$, with the results of the filter shown in Figure 5, where r is the roll output, p is the pitch output, and y is the yaw output. The RMSE is calculated according to Equation 8, comparing the estimations against the 'ideal' rotation sequence, with the results given in Figure 5:

$$RMSE = \sqrt{\sum_{i=1}^n \frac{(\hat{y}_i - y_i)^2}{n}}. \quad (8)$$

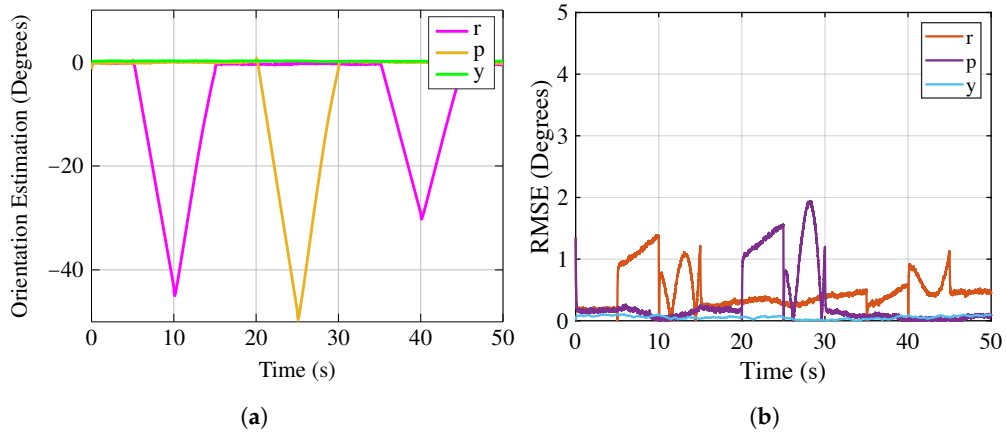


Figure 5. (a) CF orientation estimation of simulated IMU. (b) RMSE CF algorithm.

From Figure 5, only a small error deviation of less than 2° from the rotation sequence is measured, with an average RMSE of less than 1° , as shown in Table 2.

Table 2. Average RMSE results for CF against ground truth.

Roll	Pitch	Yaw
0.2863°	0.2052°	0.1481°

3.3. Mechanical Subsystem

The 3-Revolute-Revolute-Spherical (3-RRS) platform, shown in Figure 6, is designed to provide an omnirrotational structure to actuate the coffee cup. This structure is chosen for its stability, precision, and robustness against gimbal lock.

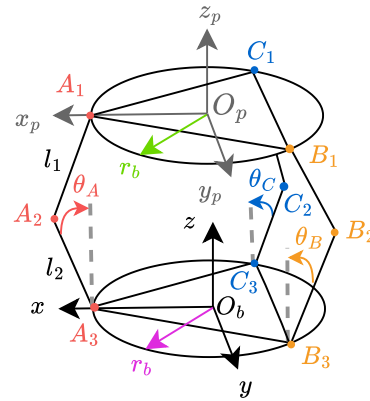


Figure 6. 3-RRS parallel manipulator diagram

Table 3. RRS symbol meanings.

Symbol	Meaning	Symbol	Meaning
A_1, B_1, C_1	Equilateral triangle points on the manipulator plate (Spherical joints)	A_3, B_3, C_3	Equilateral triangle points on the manipulator base (Revolute joints)
A_2, B_2, C_2	Mid-leg points (Revolute joints)	A, B, C	Manipulator legs
$[x_p \ y_p \ z_p]$	Plate axis	$[x \ y \ z]$	Base axis
O_p	Plate origin	O_b	Base origin
r_p	Plate radius	r_b	Base radius
$\theta_A, \theta_B, \theta_C$	Angles of the lower leg links	l_1, l_2	Upper and lower leg links

To manipulate the structure, IK analysis needs to be applied such that the co-ordinates of each joint are represented in terms of the system parameters and axes [18]. The manipulator can be decomposed into three parts: the plate, the mid-joints, and the base. For the plate, the positions of points A_1 , B_1 , and C_1 are related to the normal vector $n = [\alpha \ \beta \ \gamma]$, and the body co-ordinate system, as shown in Figure 7.

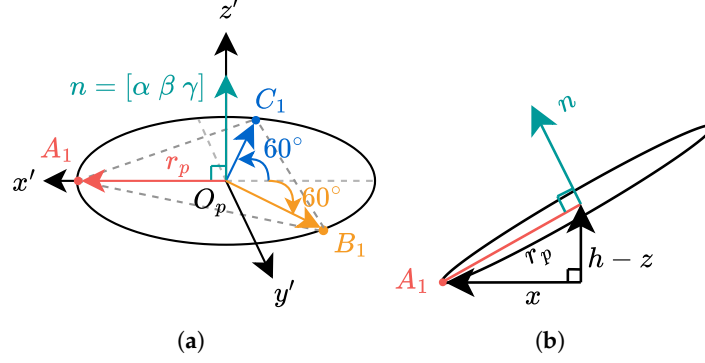


Figure 7. (a) 3-RRS parallel manipulator plate. (b) Geometric view of plate joints.

Table 4 shows the final expressions for the plate joints. The full derivation is given in the Appendix, in Section A.1.

Table 4. Expressions for 3-RRS plate joints.

Joint Co-ordinate	A_1	B_1	C_1
x	$\frac{r_p \gamma}{\sqrt{\alpha^2 + \gamma^2}}$	$-\frac{r_p \gamma}{\sqrt{\alpha - 2(3)^{-\frac{1}{2}} \alpha \beta + 3\beta^2 + 4\gamma^2}}$	$-\frac{r_p \gamma}{\sqrt{\alpha + 2(3)^{-\frac{1}{2}} \alpha \beta + 3\beta^2 + 4\gamma^2}}$
y	0	$\frac{\sqrt{3} r_p \gamma}{\sqrt{\alpha - 2(3)^{-\frac{1}{2}} \alpha \beta + 3\beta^2 + 4\gamma^2}}$	$-\frac{\sqrt{3} r_p \gamma}{\sqrt{\alpha + 2(3)^{-\frac{1}{2}} \alpha \beta + 3\beta^2 + 4\gamma^2}}$
z	$h - \frac{r_p \alpha}{\sqrt{\gamma^2 + \alpha^2}}$	$h + \frac{r_p (\alpha - \sqrt{3}\beta)}{\sqrt{\alpha - 2(3)^{-\frac{1}{2}} \alpha \beta + 3\beta^2 + 4\gamma^2}}$	$h + \frac{r_p (\alpha + \sqrt{3}\beta)}{\sqrt{\alpha + 2(3)^{-\frac{1}{2}} \alpha \beta + 3\beta^2 + 4\gamma^2}}$

Similarly, the middle joints of each leg were also computed by using the general equation for a circle, as shown in Figure 8.

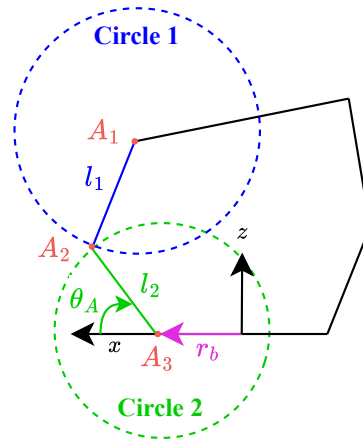


Figure 8. 3-RRS parallel manipulator middle-point diagram.

The final expressions for the mid-joints are shown in Table 5, with the full derivation given in the Appendix in Section A.2.

Table 5. Expressions for 3-RRS middle joints.

Joint Co-ordinate	A_2	B_2	C_2
x	$\frac{-D+\sqrt{D^2-4CE}}{2C}$	$\frac{-D-\sqrt{D^2-4CE}}{2C}$	$\frac{-D-\sqrt{D^2-4CE}}{2C}$
y	0	$-\sqrt{3}x$	$\sqrt{3}x$
z	$\sqrt{l_2^2 - (x - r_b)^2}$	$\sqrt{l_2^2 - 4r_b x - 4x^2 - r_b^2}$	$\sqrt{l_2^2 - 4r_b x - 4x^2 - r_b^2}$

The elevation angle between the base of the structure and each leg is computed as shown in Table 6, with the full derivation given in the Appendix in Section A.3. To achieve the desired kinematic orientation of the structure, the servo motors are actuated according to these elevation angles.

Table 6. Expressions for 3-RRS elevation angles.

Joint Angle	Expression
θ_A	$\arctan\left(\frac{z}{x-r_b}\right)$
θ_B	$\arctan\left(\frac{z}{\sqrt{x^2+y^2-r_b^2}}\right)$
θ_C	$\arctan\left(\frac{z}{\sqrt{x^2+y^2-r_b^2}}\right)$

The plate of the 3-RRS manipulator is defined within the body co-ordinate system, along the (x_p, y_p, z_p) axis in Figure 6, while the base of the structure, and the IMU, is defined within the global frame (x, y, z) . In order to specify the body frame in reference to the global frame, the Euler angle rotation matrix, given by roll (φ), pitch (θ), and yaw (ψ), angles is calculated as

$$\begin{aligned}
 {}^G_B \mathbf{R} &= \mathbf{R}_z(\psi) \mathbf{R}_y(\theta) \mathbf{R}_x(\varphi), \\
 &= \begin{bmatrix} C\psi C\theta & -S\psi C\theta + C\psi S\theta S\varphi & S\psi S\theta + C\psi S\theta C\varphi \\ S\psi C\theta & C\psi C\theta + S\psi S\theta S\varphi & -C\psi S\theta + S\psi S\theta C\varphi \\ -S\theta & C\theta S\varphi & C\theta C\varphi \end{bmatrix}, \quad (9)
 \end{aligned}$$

where $C\varphi \triangleq \cos(\varphi)$ and $S\varphi \triangleq \sin(\varphi)$. The subscript B denotes the current body reference frame (x_p, y_p, z_p) , and the superscript G denotes the destination global frame (x, y, z) . Conversion from the body to the global frame is necessary for control design, as servo actuation in the global frame produces a rotation in the body frame.

To dynamically model the system, several assumptions are made [10]:

1. There is no friction within the mechanism;
2. There is no slippage constraint within the mechanism;
3. The structure is rigid, and the plate is considered homogenous and symmetric;
4. The cup and plate are always in contact, with the liquid responding according to the pendulum dynamics (given in Section 3.4);
5. The manipulator has 3 holonomic constraints from the spherical and revolute joints;
6. Applied acceleration only occurs along the x-axis and y-axis, with a constant gravitational acceleration along the z-axis.

The Lagrangian approach is applied to obtain the dynamic model of the manipulator plate and legs. The motion equation of the system is modelled according to

$$M(q)\ddot{q} + Q(q, \dot{q}) + G(q) = \sum_{i=1}^3 J_i^T F_i, \quad (10)$$

where $q = [x \ y \ z \ \varphi \ \theta \ \psi]^T$, $Q(q, \dot{q})$ is the vector containing the Coriolis and centrifugal forces, $G(q)$ is the vector containing the gravitational forces, J_i is the Jacobian matrix, and F_i is the force on the plate applied by each leg [10].

Moreover, the leg dynamics are also derived from the Lagrangian method

$$M(q)\ddot{q} + Q(q, \dot{q})\dot{q} + G(q) = \tau + J^T F, \quad (11)$$

where in the case of leg A , $q = [\theta_{A_3} \ \theta_{A_2}]^T$ is the angle between the base and the middle joint, $\tau = [\tau \ 0]^T$ is the servomotor torque vector, $M(q)$ is the matrix of inertia, $q, \dot{q}, \ddot{q}$ is the joint position, velocity, and angular acceleration vectors, $Q(q, \dot{q})$ is the vector containing the Coriolis and centrifugal forces, and $G(q)$ is the vector containing the gravitational forces [10]. Table 7 details the parameters for the structure.

Table 7. 3-RRS model parameters.

Parameter	Value	Units
m_p	0.36	kg
m_{l1}	0.02	kg
m_{l2}	0.02	kg
l_1	0.15	m
l_2	0.7	m
I_{xx}	0.04	kg/m^2
I_{yy}	0.0693	kg/m^2
I_{zz}	3.96	kg/m^2

Simulating the manipulator in MATLAB, the SimScape toolbox can be used to create a 3-dimensional model of the structure according to the system dynamics. Figure 9 shows the stationary and dynamic motions applied to the 3-RRS model.

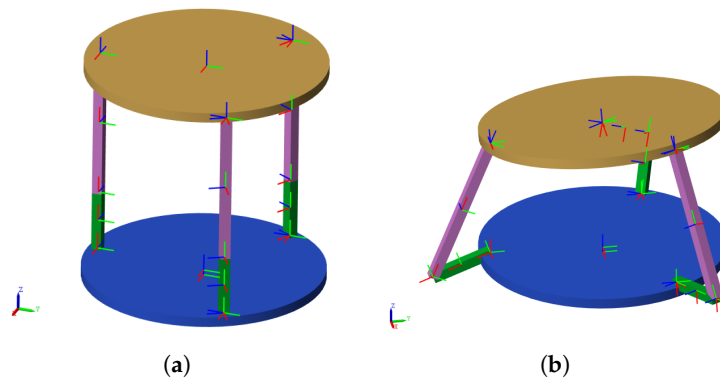


Figure 9. (a) 3-RRS stationary model in SimScape. (b) 3-RRS dynamic model in SimScape.

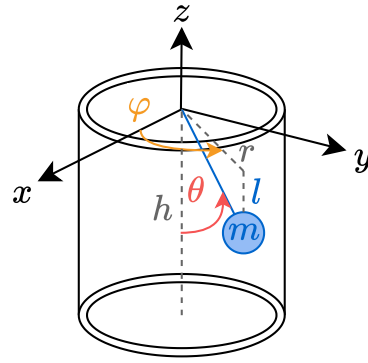
The platform was simulated using different leg-to-platform ratios, until the final dimensions given in Table 8 were reached. These dimensions allowed for a maximum tilt angle of 65° , and a servomotor range of $0 - 90^\circ$.

Table 8. 3-RRS dimensions after simulation.

Parameter	Meaning	Value	Units
r_b	Platform radius (yellow)	9	cm
r_p	Base radius (blue)	9	cm
l_1	Upper link of the leg (purple)	15	cm
l_2	Lower link of the leg (green)	7	cm
h	Height of the manipulator	13.2	cm

3.4. Control Subsystem

For liquid control, a pendulum can be used to model the dynamics of the coffee in the cup by approximating the liquid oscillations as shown in Figure 10. The variable θ is the tilt angle of the pendulum from the center axis along the elevation plane, and the symbol φ is the angle of rotation in the azimuth plane [13].

**Figure 10.** Pendulum approximation of liquid in a cup.

From the model, the natural frequency of the j^{th} sloshing mode is calculated as

$$\omega_{nj} = \sqrt{\frac{g\zeta_j}{r} \tanh\left(\frac{h\zeta_j}{r}\right)}, \quad (12)$$

where r is the cup radius, h is the height, and ζ_j is the j^{th} root of the derivative of the Bessel function of the first kind. Furthermore, the damping coefficient δ_j of the j^{th} sloshing mode in the cup is calculated as

$$\delta_j = \frac{2.89}{\pi} \left(\frac{v}{(r^3 g)^{-\frac{1}{2}}} \right)^{-\frac{1}{2}} \left(1 + \frac{(0.318)(1 - \frac{r}{h})}{\sinh(\frac{1.84h}{r}) \cosh(\frac{1.84h}{r})} \right), \quad (13)$$

where v is the viscosity of the liquid in the container. For this system, only the first asymmetric mode (δ_1) was considered, which is more dominant than the other slosh modes [13]. The system parameters are given in Table 9, from which the length of the pendulum could be calculated as $l = g/\omega_{n1}^2 = 2.3 \text{ cm}$.

Table 9. Parameters of coffee cup and liquid.

Parameter	Value	Unit
r	4.5	cm
h	10	cm
g	9.81	m^2/s
v	$1.004 \cdot 10^{-6}$	m^2/s
ζ_1	1.8412	-
ω_{n1}	20.6122	rad/s
δ_1	0.0037	-

The position of mass m in space is calculated as

$$\begin{bmatrix} x_m \\ y_m \\ z_m \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} + Rot_z(\varphi) Rot_y(-\theta) \begin{bmatrix} x + l \cos(\varphi) \sin(\theta) \\ y + l \sin(\varphi) \cos(\theta) \\ z - l \cos(\theta) \end{bmatrix}, \quad (14)$$

where the pendulum swing dynamics are linearised for small angle variations around $\theta = 0^\circ$ to obtain the linear model

$$u = \ddot{\theta} + 2\delta_1\omega_{n,1}\dot{\theta} + \omega_{n,1}^2\theta. \quad (15)$$

The platform position was found as $[x \ y \ z] = [0 \ 0 \ 13]$ according to the 3-RRS parallel manipulator design in Section 3.3 [13]. From the differential equation in 15, the slosch angles could be determined as

$$\begin{aligned} \theta &= \arctan\left(\frac{\sqrt{\ddot{x}^2 + \ddot{y}^2}}{g + \ddot{z}}\right), \\ \varphi &= \pi + \arctan\left(\frac{\ddot{y}}{\ddot{z}}\right). \end{aligned} \quad (16)$$

Following the slosch model, the controller architecture needs to track the pendulum motion, minimising the error between the target trajectory and the actual motion of the system. As shown in Figure 11, two controllers were needed, one along the roll axis, and another along the pitch axis.

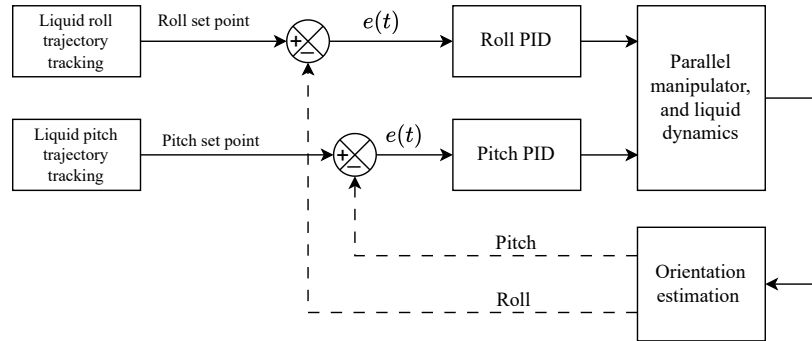


Figure 11. Full closed-loop control scheme for coffee cup stabiliser.

The continuous-time PID pitch and roll controllers are given as

$$\begin{aligned} \theta(t) &= k_{p,p}e_\theta(t) + k_{i,p}\int_0^t e_\theta(t)dt + k_{d,p}\frac{de_\theta(t)}{dt}, \\ \varphi(t) &= k_{p,r}e_\varphi(t) + k_{i,r}\int_0^t e_\varphi(t)dt + k_{d,r}\frac{de_\varphi(t)}{dt}, \end{aligned} \quad (17)$$

where subscript r indicates the roll PID terms, and subscript p indicates the pitch PID terms. Additional non-linearities exist in the system as a result of the liquid coffee and the manipulator structural mechanics. As such, MATLAB System Identification (SID) is needed to classify the overall transfer function, in addition to the dynamic model and pendulum approximation from Equations 10, 11, and 15. The SID was plotted against the captured orientation data, or validation y_1 , and the results are shown in Figure 12.

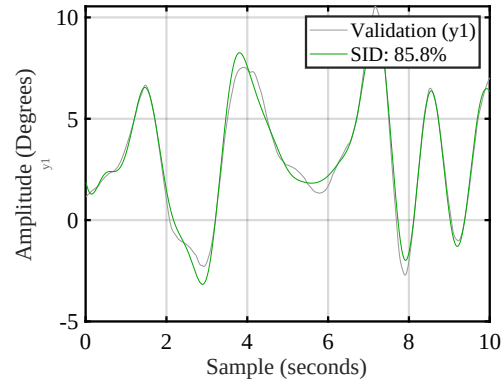


Figure 12. SID of coffee cup stabiliser system

The SID transfer function was approximated as

$$G(s) = \frac{24.22s^3 + 9.929s^2 + 30.96s - 12.96}{s^5 + 5.185s^4 + 55.17s^3 + 62.21s^2 + 146s + 5.607}. \quad (18)$$

Finally, the Zeigler-Nichols method [21] to find the proportional (K_p), integral (K_i), and derivative (K_d) terms, where

$$\begin{aligned} K_p &= k_u / 1.7, \\ K_i &= 1.2k_u / T_u, \\ K_d &= 0.075k_u T_u, \end{aligned} \quad (19)$$

is used to fine tune the PID controller according to the dynamic model and the SID transfer function. The PID controller values are shown in Table 10.

Table 10. PID controller values.

Control Type	K_p	K_i	K_d
PID	0.7128	0.0537	0

Figure 13(a) shows the simulated system response when a unit step input was applied without the controller present. The system exhibits an under-damped response, with oscillations that slowly decay over time. After introducing the PID controller, the step response results are shown in Figure 13(b) for the roll controller.

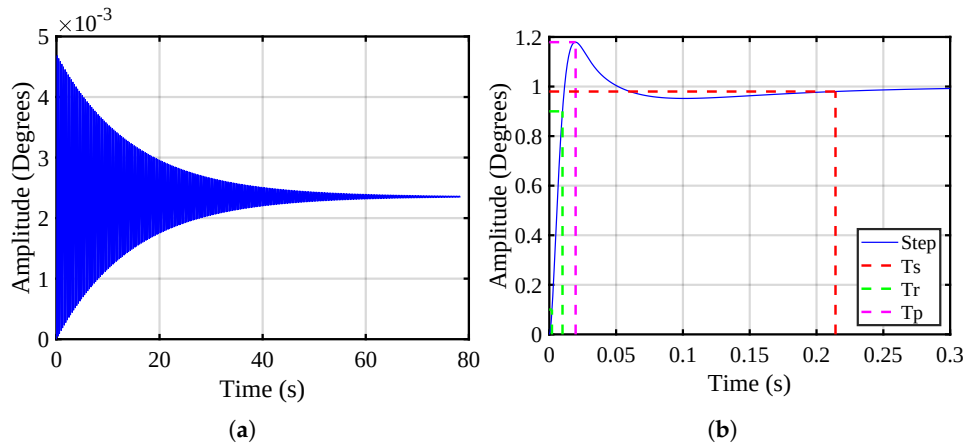


Figure 13. (a) System response to step input before PID controller. (b) System roll response to step input after PID controller.

The performance result for the pitch and roll PID controllers is given in Table 11. Since each axis is actuated independently, the same controller can be implemented in each direction and fine-tuned accordingly.

Table 11. PID closed-loop system response for step input pitch and roll.

Controller	T_r (s)	T_s (s)	T_p (s)
PID	0.0078	0.2142	0.0197

3.5. Integrated System

The final system implementation shown in Figure 14 includes an STM32 H743ZI-Nucleo board for the system control (Section 3.4) and orientation estimation algorithm (Section 3.2), two LSM6DSOX IMUs for the accelerometer and gyroscope data capture, servomotors for actuation, a power supply, and a custom 3D-printed 3-RRS parallel manipulator, designed according to the simulated system constraints (Section 3.3).

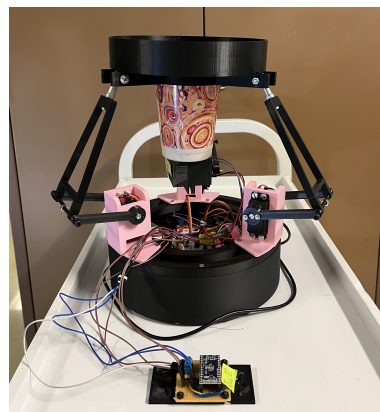


Figure 14. Integrated system

4. Results

This section presents the results of the experimental testing conducted on the coffee-stabiliser. Under typical driving conditions, the system must maintain the resultant acceleration vector at an angle that is approximately perpendicular to the base of the coffee cup, to prevent coffee spillage. This is demonstrated by the system's ability to track the 'reference' vector that directly opposes the applied roll, and pitch movements on the cup and liquid inside.

4.1. Experimental Set-up

Firstly, the coffee-cup stabiliser was tested in laboratory conditions, under a controller environment, whereby the system was placed on a trolley, and the braking, turning, accelerating, and tilting motions were applied by hand. The maximum applied G-force was set at 0.35 G, which approximates a very sharp change in acceleration under driving conditions. The coffee stabiliser needs to be placed with servomotor 1 on the imaginary pitch axis, aligned with the longitudinal axis of the trolley, as shown in Figure 15, to keep track of the direction in which the accelerations are applied to the system. Thereafter, the system needs to be powered by connecting the battery. A bluetooth module was used to stream the data transmission, enabling the applied accelerations, the cup's orientation, and the resultant acceleration data to be collected and interpreted.

4.2. Workflow Generation

The testing process to generate results is shown below:

1. Move the platform rapidly from left to right along the longitudinal (pitch) axis, varying the accelerations from 0 - 0.35 G; 299
2. Return platform to starting position; 300
3. Move the platform rapidly from left to right along the lateral (roll) axis, varying the accelerations from 0 - 0.35 G; 301
4. Return platform to starting position; 302
5. Plot results in MATLAB; 303
6. Repeat for a total of 5 tests. 304

4.3. Results 307

4.3.1. Laboratory Test Conditions 308

Figure 15 shows the pitch and roll tilt testing being performed on the system, as well as sudden accelerations left and right applied under laboratory conditions. 309

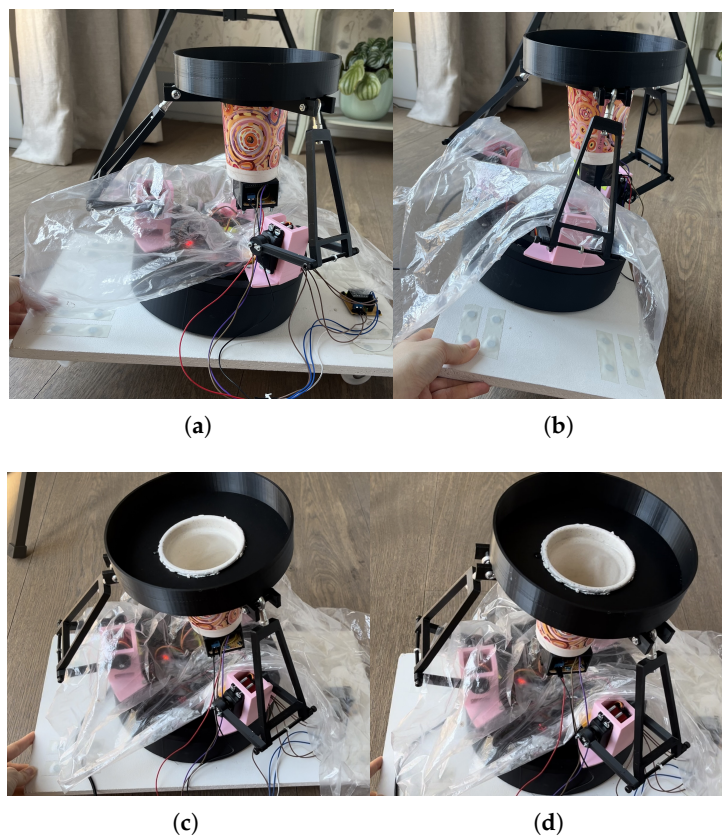


Figure 15. System response under laboratory conditions: (a) Left pitch tilt, (b) Right roll tilt, (c) Sudden acceleration right, (d) Sudden acceleration left. 311

The resultant vector on the base of the coffee cup is shown in Figure 16, where the 'reference' value is the required vector needed in order to cancel out the applied accelerations on the system, and the 'body' is the cup's actual position from the applied accelerations. This was decomposed along each axis. 312
313
314

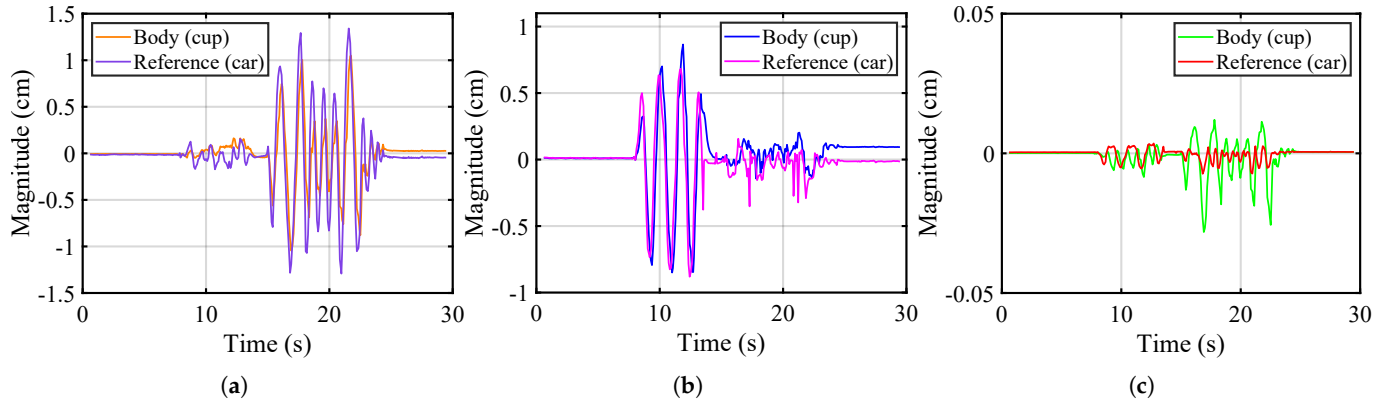


Figure 16. System tracking of body (cup) vector and reference (car) vector along different axes: (a) x-axis, (b) y-axis, (c) z-axis.

The change along the x-, y-, and z- axes corresponds to the pitch and roll of the body vector. This is shown in the pitch and roll plots in Figure 17.

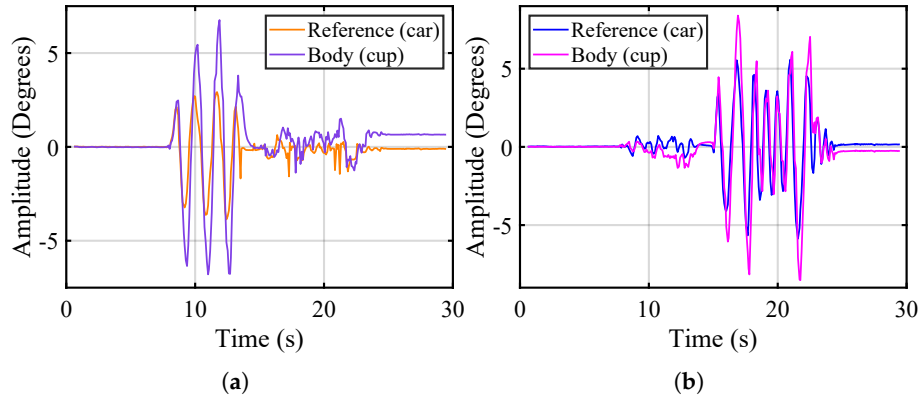


Figure 17. Pitch and roll of body vector compared to 'reference' vector: (a) Pitch axis, (b) Roll axis.

The test was repeated several times to create an average absolute error result. The calculation is given as

$$e_{ave} = \frac{1}{N} \sum_{i=1}^N |\hat{d}_i - d_i|, \quad (20)$$

where N is the total number of samples, $\hat{d}$ is the predictive value at sample i , and d is the true value. The results from five tests are summarised in Table 12.

Table 12. Summary of absolute error results of plots (°).

Test	Absolute error roll plot (°)	Absolute error pitch plot (°)	Maximum absolute error roll plot (°)	Maximum absolute error pitch plot (°)
1	0.7503	0.7318	4.8052	5.2018
2	0.6030	0.4457	3.9879	4.5767
3	0.8140	0.6433	5.9624	4.1339
4	0.5178	1.0706	5.6107	4.0793
5	0.8964	0.9252	5.1342	4.2502
Average	0.7162	0.7633	5.1001	4.484

A histogram was plotted to evaluate the angle error between the target ‘reference’ vector roll and pitch, and the system roll and vector response. The histograms are shown in Figure 18.

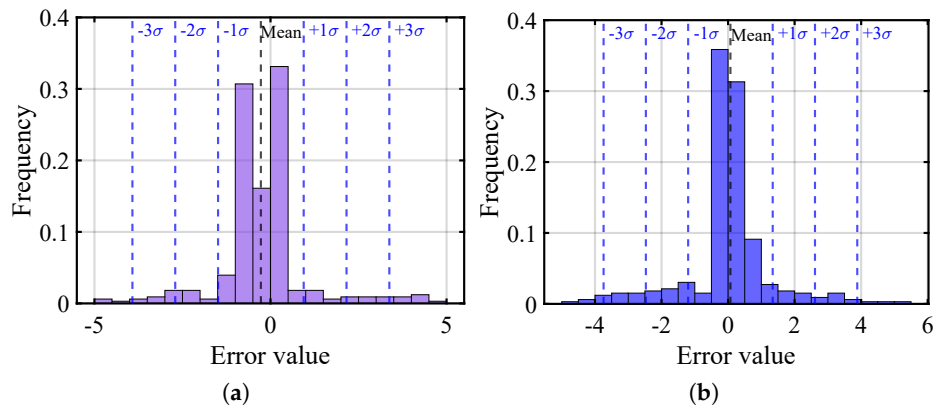


Figure 18. Histograms of errors between ‘reference’ and body vectors: (a) Pitch axis, (b) Roll axis.

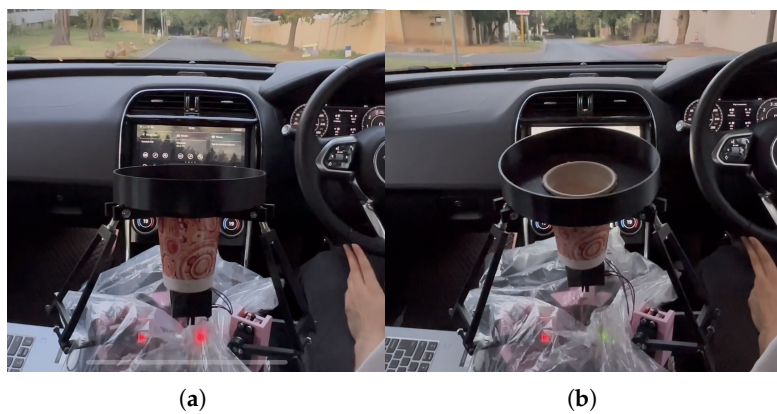
The histograms for each test were plotted, and the results of the mean and standard deviation for five tests is given in Table 13.

Table 13. Summary of mean and standard deviation results of error histogram plots.

Test	Mean roll plot (°)	Standard deviation roll plot (°)	Mean pitch plot (°)	Standard deviation pitch plot (°)
1	−0.2733	1.2161	0.0708	1.2708
2	−0.2488	0.8262	−0.0617	0.9560
3	0.08923	1.2764	−0.1443	1.0291
4	−0.0446	1.0418	−0.07719	1.5130
5	−0.2105	1.4792	0.0602	1.3108
Average	−0.1376	1.1679	−0.0304	1.2159

4.3.2. Field Test Conditions

To assess the system’s performance under real-world driving conditions, the stabiliser was placed in a moving vehicle and underwent turning, braking, and accelerating motions, as shown in Figure 19.



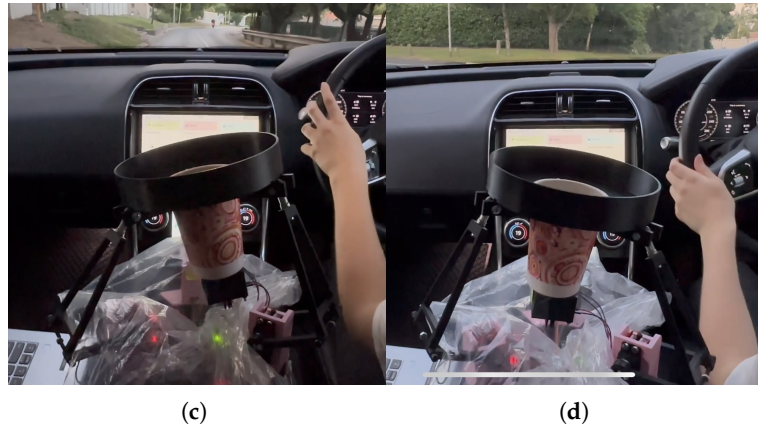


Figure 19. System response during real-world driving conditions: (a) Downhill acceleration, (b) Sudden braking, (c) Left turning, (d) Right turning.

A segment of the driving test was recorded, as shown in Figures 20 and 21, whereby the stabiliser responded to the car braking and accelerating along a smooth, flat part of the road, with minimal mechanical and surface vibrations.

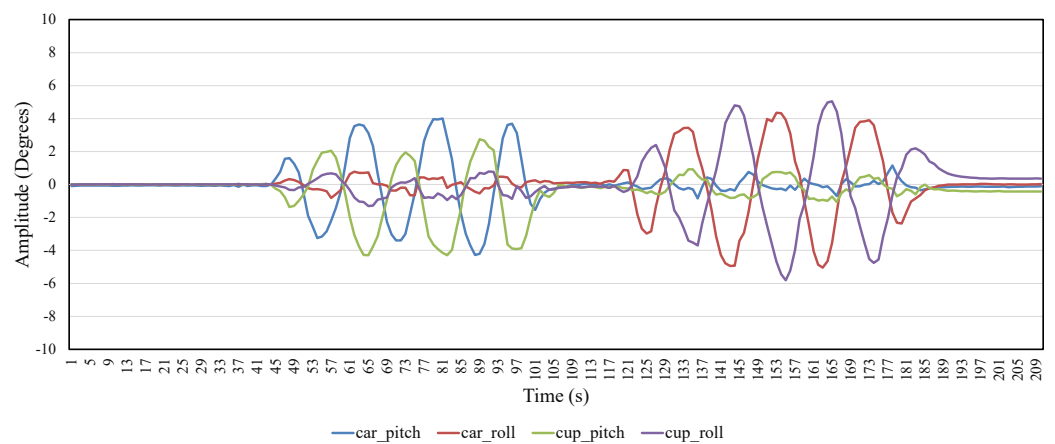


Figure 20. Roll and pitch response on smooth road.

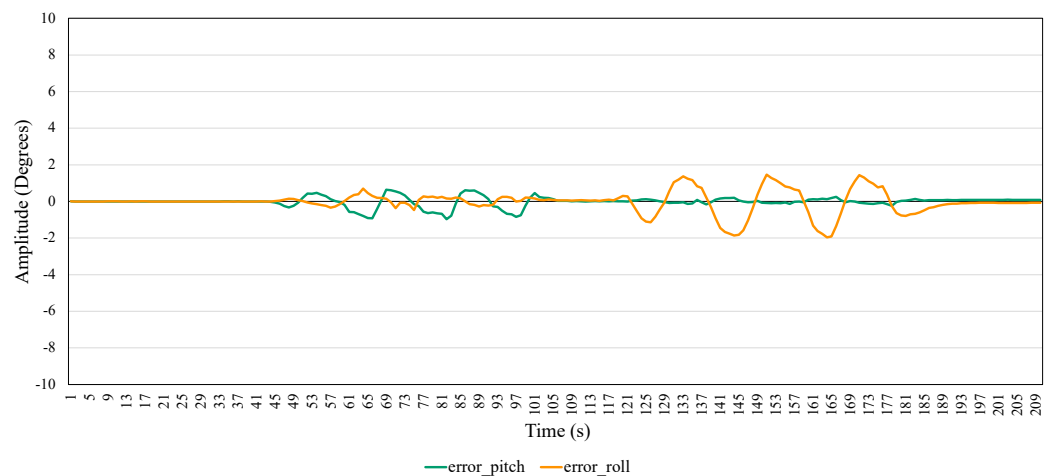


Figure 21. Roll and pitch errors on smooth road.

When tested on a rough road surface, with potholes and obstructions, additional vibrations were added to the system, and erratic motion was recorded, as shown in Figure 22 and 23.

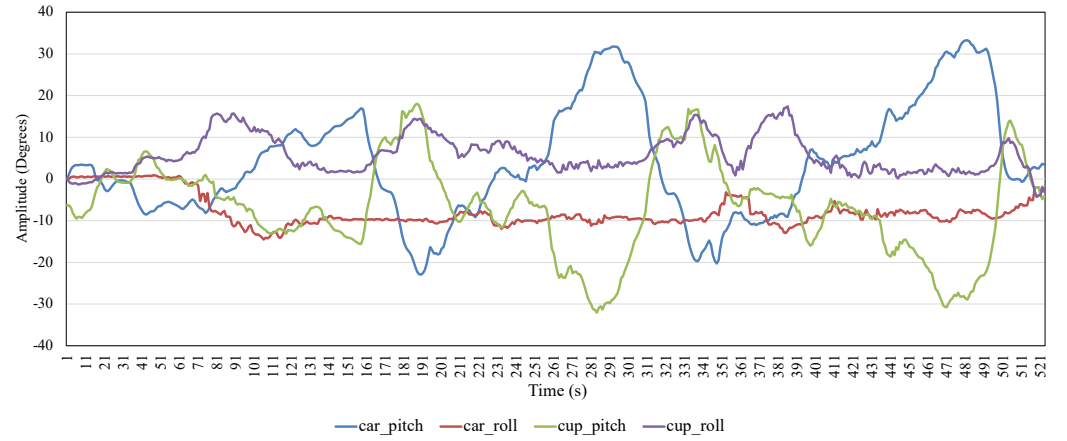


Figure 22. Roll and pitch response on rough road.

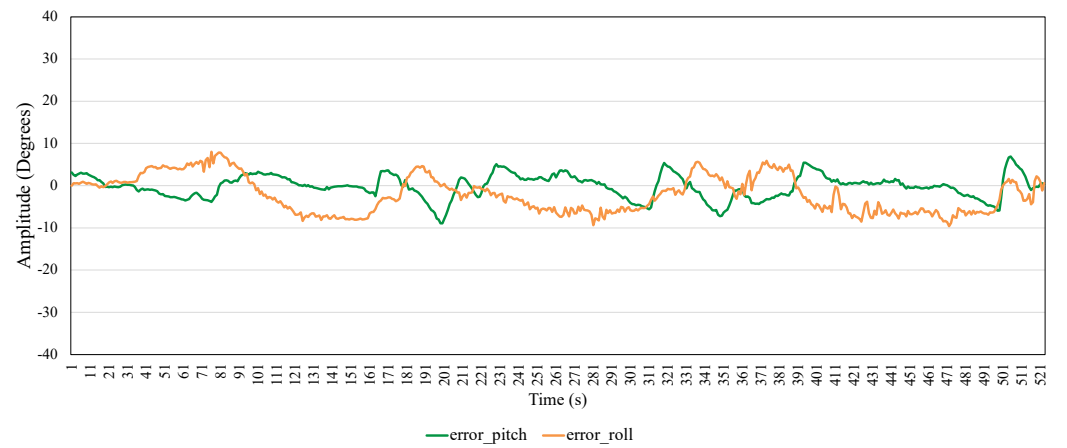


Figure 23. Roll and pitch errors on rough road.

5. Discussion

5.1. Analysis of Results

This work proposes an omnirotational platform in the form of a 3-RRS parallel manipulator, in conjunction with a PID controller and CF orientation estimation algorithm, for liquid control in a moving vehicle. Overall, the experimental results demonstrated the system's ability to prevent coffee spillage by actuating the coffee cup against the applied accelerations.

Under controlled laboratory conditions on a moving trolley, with gradual changes applied, the body vector very closely approximated the target reference vector, with the x-axis, y-axis, and z-axis motions producing the roll and pitch results in Figure 17. Small errors were observed, as shown in Table 12, and from the statistical analysis in the histogram plots, the data clustered around the 0-value for the angle errors.

The controlled laboratory results are similar to that of Yano et. al [1] when H^∞ control was used for slosh suppression and the water displacement was observed at 0.5 cm to 1 cm, similar to the vector response in Figure 16. When paired with a transfer rig tuned to the applied input frequency, Yano et. al [1] achieved complete slosh suppression, which was not fully reached in the case of this study.

The plug-in feed-forward control presented by Biagiotti et. al [13] produced a sloshing angle of approximately 3° when the natural frequency and damping ratio of the liquid was known, which is similar to the results in Figure 17. In Chen et. al [15], the tilting angle deviation was greatly improved with acceleration compensation, resulting in a maximum displacement of 2.857 mm. The slosh control in this paper however produced a slightly

larger displacement, shown in Table 12, but was equally able to maintain the liquid level below the cup rim.

When tested in real-world driving conditions on smooth road surfaces, as shown in Section 4.3.2, the system was able to closely track the ‘reference’ vector and control the liquid in the cup, as shown in Figure 20. Only a small angle deviation was observed in Figure 21. As more road vibrations and irregular driving motions were introduced in Figure 22, the error deviation was larger, as shown in Figure 23. Nonetheless, the system maintained the liquid level within the coffee cup by applying opposing roll and pitch motions without any spillages occurring.

This study, in comparison to the laboratory work of Yano et. al [1], Biagiotti et. al [13], and Chen et. al [15], offers a more general, omnirotational platform, with the objective of suppressing a variety of sloshing frequencies and unknown non-linearities caused by on-road conditions. It was concluded that for gradual driving motions (turning, braking and accelerating), the system was able to closely track the resultant acceleration vector closely, preventing spillage, however for instantaneous accelerations from potholes or large road irregularities, the system could not fully suppress the sloshing motion, only reduce it. Few real-world experiments exist of liquid control in this manner in a moving vehicle, and thus this study offers a new perspective in the field.

5.2. Future Work

Future work will explore the use of non-invertible control models for more robust design and aggressive controller action against instantaneous accelerations.

While PID control offers a trade-off between performance and achievable actuator authority, Sliding Mode Control (SMC) [9] and Model Predictive Control (MPC) are suitable alternative design options, however, a more accurate system model would be needed for simulation. The driving surface and mechanical vibrations (such as engine idling, changing gears, suspension, potholes, and rough road conditions) would need to be identified as part of the design, as well as an improved liquid model, since the pendulum slosh approximation does not consider higher modes of oscillation in the liquid container.

Given the complexity of this multi-variable modelling for traditional control approaches, using a neural network (NN) offers another alternative. A machine-learning (ML) algorithm could be trained on typical driving scenarios, producing a dynamic model to be used in reference tracking. In this way, ‘black box’ dynamics can improve tracking and actuation accuracy of the coffee-sup stabiliser, however, the computational complexity of the system would increase, requiring more powerful hardware.

6. Conclusion

This study presented the implementation of a 3-RRS parallel manipulator as a coffee-cup stabiliser and anti-spill device, demonstrating an effective approach to liquid control in a mobile environment. The primary objective of the system was to track the resultant acceleration vector (a_{res}) and apply a system vector (a_{sys}), with corresponding roll and pitch motions, ensuring a_{res} was maintained at an angle perpendicular to the base of the cup.

Three components were needed to realise the stabiliser system: a measurement unit, a control algorithm, and a mechanical structure. The measurement unit consisted of an IMU and a CF algorithm, for sensor fusion of the accelerometer and gyroscope readings. The control algorithm comprised of a PID controller combined with trajectory tracking, to reach the orientation set-point of the liquid in space. These components were housed within a 3D-printed 3-RRS parallel manipulator mechanical structure for actuation of the coffee cup.

The system was tested in both laboratory and field conditions. Placing the stabiliser on a trolley and manually applying various motions, the system was shown to closely track the

‘reference’ vector, with very minor angle deviations. These results were equivalent to other examples of liquid control in literature. When tested under real-world driving conditions, the system was effective at preventing coffee spillage on smooth road conditions and under gradual driving motions, such as turning, braking and accelerating, but was unable to completely suppress sloshing during instantaneous movement, such as from potholes and asphalt irregularities.

The system nonetheless offers an effective, omnirotational approach to liquid control in a moving vehicle, laying the foundation for further research using this approach.

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Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

3-RRS	3-Revolute-Revolute-Spherical
3-RPS	3-Revolute-Prismatic-Spherical
RUU	Revolute-Universal-Universal
MEMS	Micro-electromechanical Sensor
CF	Complimentary Filter
PID	Proportional-Integral-Derivative
IK	Inverse Kinematic
IMU	Inertial Measurement Unit
KF	Kalman Filter
EDG	Estimation by Gradient Descent
FL	Fuzzy Logic
DOF	Degrees-of-freedom
LQI	Linear Quadratic Integration
LPF	Low-pass Filter
SID	System Identification
SMC	Sliding Mode Control
MPC	Model Predictive Control
NN	Neural Network
ML	Machine Learning
SBC	Single-board Computer

Appendix A

The derivation of the IK expressions for the 3-RRS parallel manipulator are given here, adapted from [20] according to the coffee-cup stabiliser geometric structure:

Appendix A.1

The plate points A_1 , B_1 and C_1 need to be expressed in terms of the manipulator parameters and the normal vector, as shown in Figure 7.

Taking point A_1 as an example, the dot product was used between $A_1 = (x, y, z - h)$ and the normal vector as

$$\langle x, y, (z - h) \rangle \cdot \langle \alpha, \beta, \gamma \rangle = 0. \quad (\text{A1})$$

Since A_1 was located on the x-axis, $y = 0$, thus the equation was simplified to

$$\alpha x + (z - h)\gamma = 0. \quad (\text{A2})$$

From the geometry in Figure 7, A_1 could also be represented using Pythagoras' Theorem, as

$$\begin{aligned} r_p^2 &= x^2 + (h - z)^2, \\ x &= \pm \sqrt{r_p^2 - (h - z)^2}. \end{aligned} \quad (\text{A3})$$

Recalling that A_1 needed a positive x value, as it was located on the positive quadrant of the y-axis, z could be solved for as

$$\alpha \sqrt{r_p^2 - (h - z)^2} + \gamma(z - h) = 0, \quad (\text{A4})$$

which after isolating the z term was found as

$$\begin{aligned} r_p^2 &= \frac{\gamma^2}{\alpha^2}(z - h)^2 + (h - z)^2, \\ (z - h) &= \pm \sqrt{\frac{r_p^2 \alpha^2}{\gamma^2 + \alpha^2}}, \\ z &= h \pm \frac{r_p \alpha}{\sqrt{\gamma^2 + \alpha^2}}. \end{aligned} \quad (\text{A5})$$

To find x , z from Equation A5 was substituted into Equation A3, such that

$$\begin{aligned} x &= \sqrt{r_p^2 - \left(h - \left(h - \frac{r_p \alpha}{(\gamma^2 + \alpha^2)^{-\frac{1}{2}}} \right) \right)^2}, \\ &= \frac{r_p \gamma}{\sqrt{\alpha^2 + \gamma^2}}. \end{aligned} \quad (\text{A6})$$

Equation A6 concluded the derivation of the A_1 co-ordinates, as x , y , and z were represented in terms of the normal vector and manipulator parameters. The final results are given in Table 4. The same procedure was applied to derive the joints B_1 and C_1 .

Appendix A.2

The joint points A_2 , B_2 and C_2 need to be expressed in terms of the manipulator parameters, as shown in Figure 8. Two circles are used, with radii l_1 and l_2 to solve for the unknown joint co-ordinates.

Taking point A_2 as an example, $A_2 = (x, y, z)$ is located on the circumference of circle 1 and the circumference of circle 2 according to Equation A7 and A8 respectively:

$$(x - x_{A_1})^2 + (y - y_{A_1})^2 + (z - z_{A_1})^2 = l_1^2 \quad (\text{A7})$$

$$(x - x_{A_3})^2 + (y - y_{A_3})^2 + (z - z_{A_3})^2 = l_2^2 \quad (\text{A8})$$

Noting that $A_3 = (r_b, 0, 0)$, located on the $y = 0$ axis, as shown in Figure 8, Equation A8 could be simplified further as

$$\begin{aligned} l_2^2 &= (x - r_b)^2 + (y - 0)^2 + (z - 0)^2, \\ z &= \pm \sqrt{l_2^2 - (x - r_b)^2}. \end{aligned} \quad (\text{A9})$$

Expanding Equation A7, and substituting z from Equation A9,

$$\begin{aligned} l_1^2 &= x^2 - 2x_{A_1}x + x_{A_1}^2 + z^2 - 2z_{A_1}z + z_{A_1}^2, \\ l_1^2 &= x^2 - 2x_{A_1}x + x_{A_1}^2 + (l_2^2 - (x - r_b)^2) - 2z_{A_1}\sqrt{l_2^2 - (x - r_b)^2} + z_{A_1}^2, \\ \sqrt{l_2^2 - (x - r_b)^2} &= \frac{2x(r_b - x_{A_1}) + x_{A_1}^2 + z_{A_1}^2 + l_2^2 - l_1^2 - r_b^2}{2z_{A_1}}, \\ \sqrt{l_2^2 - (x - r_b)^2} &= \frac{2x(r_b - x_{A_1})}{2z_{A_1}} + \frac{x_{A_1}^2 + z_{A_1}^2 + l_2^2 - l_1^2 - r_b^2}{2z_{A_1}}. \end{aligned} \quad (\text{A10})$$

Equation A10 was separated into

$$A = \frac{2x(r_b - x_{A_1})}{2z_{A_1}}, \quad (\text{A11})$$

and

$$B = \frac{x_{A_1}^2 + z_{A_1}^2 + l_2^2 - l_1^2 - r_b^2}{2z_{A_1}}, \quad (\text{A12})$$

for simplicity. Expanding Equation A10 with Equation A11 and A12 produced

$$\begin{aligned} l_2^2 - x^2 + 2xr_b - r_b^2 &= A^2x^2 + 2ABx + B^2, \\ 0 &= (A^2 + 1)x^2 + 2x(AB - r_b) + B^2 + r_b^2 - l_2^2, \end{aligned} \quad (\text{A13})$$

which is the standard form of a quadratic equation. Equation A13 could be further separated into

$$C = A^2 + 1, \quad (\text{A14})$$

$$D = 2(AB - r_b), \quad (\text{A15})$$

and

$$E = B^2 + r_b^2 - l_2^2. \quad (\text{A16})$$

Using the quadratic formula, x was computed as

$$x = \frac{-D \pm \sqrt{D^2 - 4CE}}{2C}. \quad (\text{A17})$$

Once x was calculated, the value of z could be found by substituting x into Equation A8. Equation A17 concluded the calculation of the middle A_2 joint. A similar process was applied to find the middle joints B_2 and C_2 . The final co-ordinate representation for the middle joints is shown in Table 5.

Appendix A.3

Taking θ_A as an example, with x , y , and z of joint A_2 parameterised, the elevation angle could be calculated using the trigonometric equations

$$\begin{aligned}
z &= l_2 \sin(\theta_A), \\
x &= r_p + l_2 \cos(\theta_A), \\
\theta_A &= \arctan\left(\frac{\sin(\theta_A)}{\cos(\theta_A)}\right),
\end{aligned} \tag{A18}$$

which produced the result

$$\theta_A = \arctan\left(\frac{z}{x - r_b}\right). \tag{A19}$$

A similar process was applied to find θ_B and θ_C . The final angle rotations to be performed by the servomotors are given in Table 6.

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